

Lecture 8: Vector fields and orientations

Suppose we have a parametrized surface

$$f: U \hookrightarrow \mathbb{R}^3 \tag{1}$$

where $U \subset \mathbb{R}^2$ is an open set with coordinates u, v .

Relative to the coordinate basis (f_u, f_v) , the metric is represented by the matrix of their pairwise inner products. Writing

$$E = \langle f_u, f_u \rangle, \quad F = \langle f_u, f_v \rangle, \quad G = \langle f_v, f_v \rangle,$$

and using $\langle f_v, f_u \rangle = \langle f_u, f_v \rangle$, we obtain the symmetric matrix of the first fundamental form:

$$g = \begin{pmatrix} E & F \\ F & G \end{pmatrix} = (Df)^\top Df.$$

For $\xi = (a, b)$,

$$I_u(\xi, \xi) = Ea^2 + 2Fab + Gb^2.$$

One often writes the same quadratic form symbolically as

$$ds^2 = E du^2 + 2F du dv + G dv^2.$$

The symbols du, dv record components of tangent vectors; they are not small finite displacements.

The immersion condition appears directly in the first fundamental form. By the cross-product identity,

$$\|f_u \times f_v\|^2 = \|f_u\|^2 \|f_v\|^2 - \langle f_u, f_v \rangle^2.$$

Thus

$$\det g = EG - F^2 = \|f_u \times f_v\|^2 > 0.$$

Consequently,

$$E > 0, \quad G > 0, \quad EG - F^2 > 0.$$

The area of the parallelogram spanned by f_u and f_v is

$$\sqrt{EG - F^2}.$$

This leads to the surface-area element

$$dA = \sqrt{EG - F^2} \, du \, dv.$$

Let $\gamma(t) = (u(t), v(t))$ be a curve in the parameter domain and let

$$c(t) = f(\gamma(t))$$

be the corresponding curve on the surface. Then

$$\dot{c} = f_u \dot{u} + f_v \dot{v}.$$

Therefore

$$\|\dot{c}(t)\|^2 = E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2 = I_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}).$$

The length of c is consequently

$$L(c) = \int_a^b \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2} dt.$$

Thus the first fundamental form contains exactly the information needed to measure lengths of curves on the surface.

The first fundamental form also measures angles. For nonzero tangent vectors X, Y ,

$$\cos \theta = \frac{I(X, Y)}{\sqrt{I(X, X)}\sqrt{I(Y, Y)}}.$$

In particular, the coordinate curves meet orthogonally precisely when

$$F = \langle f_u, f_v \rangle = 0.$$

Remarks.

- The first fundamental form depends only on first derivatives of the parametrization.
- It records lengths, angles, and areas within the surface.
- It is not generally the standard inner product on the parameter domain U .
- Its coefficients E, F, G depend on the chosen coordinates, although the geometry they describe does not.

For the graph

$$f(u, v) = (u, v, h(u, v)),$$

we found

$$f_u = (1, 0, h_u), \quad f_v = (0, 1, h_v).$$

Hence

$$E = 1 + h_u^2, \quad F = h_u h_v, \quad G = 1 + h_v^2.$$

The first fundamental form is

$$ds^2 = (1 + h_u^2) du^2 + 2h_u h_v du dv + (1 + h_v^2) dv^2.$$

Even though the parameter domain is flat, the graph generally stretches lengths in a way that depends on the derivatives of h .

Example: the saddle. For the embedded graph $f(u, v) = (u, v, uv)$,

$$f_u = (1, 0, v), \quad f_v = (0, 1, u).$$

Taking the three inner products gives

$$E = 1 + v^2, \quad F = uv, \quad G = 1 + u^2.$$

Thus

$$ds^2 = (1 + v^2) du^2 + 2uv du dv + (1 + u^2) dv^2,$$

and

$$EG - F^2 = 1 + u^2 + v^2 > 0.$$

For instance, at $f(1, 1) = (1, 1, 1)$, the coordinate tangent vectors are $(1, 0, 1)$ and $(0, 1, 1)$. Each has length $\sqrt{2}$, and

$$\cos \alpha = \frac{F}{\sqrt{EG}} = \frac{1}{2}.$$

They meet at angle $\pi/3$, even though the parameter axes meet at a right angle.

The tangent vector with coordinate components $(1, -1)$ is $f_u - f_v = (1, -1, 0)$. Its squared length is

$$E - 2F + G = 2,$$

agreeing with its Euclidean squared length in $\mathbb{R}^3$.

Example: sphere

Consider the unit sphere parametrized by longitude and latitude:

$$f(\varphi, \vartheta) = (\cos \varphi \cos \vartheta, \sin \varphi \cos \vartheta, \sin \vartheta).$$

Take $-\pi < \varphi < \pi$ and $-\pi/2 < \vartheta < \pi/2$ for an embedded coordinate patch. Differentiating gives

$$\begin{aligned} f_\varphi &= (-\sin \varphi \cos \vartheta, \cos \varphi \cos \vartheta, 0), \\ f_\vartheta &= (-\cos \varphi \sin \vartheta, -\sin \varphi \sin \vartheta, \cos \vartheta). \end{aligned}$$

Their inner products are

$$E = \cos^2 \vartheta, \quad F = 0, \quad G = 1.$$

Thus

$$ds^2 = \cos^2 \vartheta d\varphi^2 + d\vartheta^2.$$

A change $d\varphi$ in longitude has physical length $\cos \vartheta |d\varphi|$. Circles of latitude shrink toward the poles.

At $\vartheta = \pm\pi/2$, the coefficient E vanishes and the parametrization is not an immersion. The sphere is smooth there; this particular coordinate system is singular.

Example: the cylinder. For the cylinder $x^2 + y^2 = R^2$, use the embedded patch

$$f(\theta, z) = (R \cos \theta, R \sin \theta, z), \quad -\pi < \theta < \pi, \quad z \in \mathbb{R}.$$

This patch omits one vertical line. Other angular intervals give patches covering that line as well. The coordinate tangent vectors are

$$f_\theta = (-R \sin \theta, R \cos \theta, 0), \quad f_z = (0, 0, 1).$$

Consequently,

$$E = R^2, \quad F = 0, \quad G = 1, \quad ds^2 = R^2 d\theta^2 + dz^2.$$

Putting $\ell = R\theta$ gives $ds^2 = d\ell^2 + dz^2$, the Euclidean metric on a strip. Thus unrolling a cylinder preserves lengths and angles locally, although the cylinder bends in $\mathbb{R}^3$.

For a concrete length calculation, consider the helix

$$c(t) = (R \cos t, R \sin t, bt).$$

Its parameter velocity is $(\dot{\theta}, \dot{z}) = (1, b)$, so

$$\|\dot{c}(t)\|^2 = R^2 + b^2, \quad L(c|_{[0, 2\pi]}) = 2\pi\sqrt{R^2 + b^2}.$$

One full turn passes through more than one coordinate patch; the same local formula computes its length throughout.

Example: a cone without its vertex. Fix $a > 0$. The upper cone

$$S = \{(x, y, z) : z = a\sqrt{x^2 + y^2}, z > 0\}$$

is an embedded surface: it is a smooth graph over $\mathbb{R}^2 \setminus \{0\}$. A polar coordinate patch is

$$f(r, \theta) = (r \cos \theta, r \sin \theta, ar), \quad r > 0, \quad -\pi < \theta < \pi.$$

Since

$$f_r = (\cos \theta, \sin \theta, a), \quad f_\theta = (-r \sin \theta, r \cos \theta, 0),$$

we get

$$E = 1 + a^2, \quad F = 0, \quad G = r^2,$$

and therefore

$$ds^2 = (1 + a^2) dr^2 + r^2 d\theta^2, \quad dA = r\sqrt{1 + a^2} dr d\theta.$$

A radial segment from r_0 to $r_1 > r_0$ has length $\sqrt{1 + a^2}(r_1 - r_0)$; a horizontal circle has circumference $2\pi r$.

Adding the vertex does not give a smooth surface there. The ray directions $(\cos \theta, \sin \theta, a)$ cannot all lie in one two-dimensional tangent plane at the vertex.

We now verify that the first fundamental form is independent of the chosen parametrization.

Lemma. Let

$$\tilde{f} = f \circ \phi, \quad \phi : \tilde{U} \longrightarrow U$$

be a change of coordinates. If g and $\tilde{g}$ are the matrices of the first fundamental forms of f and $\tilde{f}$, then

$$\boxed{\tilde{g}(q) = D\phi(q)^{\top} g(\phi(q)) D\phi(q).}$$

Equivalently, for $\xi, \eta \in \mathbb{R}^2$,

$$\tilde{I}_q(\xi, \eta) = I_{\phi(q)}(D\phi_q \xi, D\phi_q \eta).$$

The coefficients change, but the inner product on geometric tangent vectors does not.

Proof. The chain rule gives

$$D\tilde{f}_q = Df_{\phi(q)}D\phi_q.$$

Therefore

$$\begin{aligned}\tilde{\mathbf{I}}_q(\xi, \eta) &= \langle D\tilde{f}_q\xi, D\tilde{f}_q\eta \rangle \\ &= \langle Df_{\phi(q)}D\phi_q\xi, Df_{\phi(q)}D\phi_q\eta \rangle \\ &= \mathbf{I}_{\phi(q)}(D\phi_q\xi, D\phi_q\eta).\end{aligned}$$

In matrix notation,

$$\begin{aligned}\tilde{g} &= (D\tilde{f})^\top D\tilde{f} \\ &= (D\phi)^\top (Df)^\top Df (D\phi) \\ &= (D\phi)^\top g(D\phi).\end{aligned}$$

This proves the lemma.

Example: the paraboloid in two coordinate systems. Return to the paraboloid from the earlier example,

$$f(u, v) = (u, v, u^2 + v^2).$$

The graph formula gives

$$g(u, v) = \begin{pmatrix} 1 + 4u^2 & 4uv \\ 4uv & 1 + 4v^2 \end{pmatrix}.$$

On the portion $u > 0$, introduce polar coordinates:

$$\tilde{f}(r, \theta) = (r \cos \theta, r \sin \theta, r^2), \quad r > 0, \quad -\pi/2 < \theta < \pi/2.$$

Direct differentiation gives

$$\tilde{f}_r = (\cos \theta, \sin \theta, 2r), \quad \tilde{f}_\theta = (-r \sin \theta, r \cos \theta, 0).$$

Thus

$$\tilde{E} = 1 + 4r^2, \quad \tilde{F} = 0, \quad \tilde{G} = r^2,$$

and

$$ds^2 = (1 + 4r^2) dr^2 + r^2 d\theta^2.$$

The polar coordinate curves are orthogonal. In Cartesian coordinates, the cross term $F = 4uv$ is usually nonzero. These are different coordinate grids on the same surface.

Verifying the coordinate-change formula.

For $\phi(r, \theta) = (r \cos \theta, r \sin \theta)$, set

$$A = D\phi = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad w = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}.$$

The Cartesian metric, evaluated at $\phi(r, \theta)$, is

$$g \circ \phi = I_2 + 4ww^\top.$$

The columns of A are orthogonal, and direct multiplication gives

$$A^\top A = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad A^\top w = \begin{pmatrix} r \\ 0 \end{pmatrix}.$$

Hence the lemma predicts

$$\begin{aligned} \widetilde{g} &= A^\top (I_2 + 4ww^\top) A \\ &= \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} + 4 \begin{pmatrix} r^2 & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 + 4r^2 & 0 \\ 0 & r^2 \end{pmatrix}, \end{aligned}$$

exactly as in the direct computation.

In particular, a vector with polar components (a, b) has squared length $(1 + 4r^2)a^2 + r^2b^2$. Its Cartesian components are $A(a, b)^\top$; using g gives the same squared length.

We justify the “area element” introduced earlier.

Lemma. Surface integrals with respect to the area element are independent of the parametrization.

Let $f : U \rightarrow S \subset \mathbb{R}^3$ be an embedding onto a surface patch, and let $\phi : \tilde{U} \rightarrow U$ be a diffeomorphism. Write $\tilde{f} = f \circ \phi$, with metric matrices g and $\tilde{g}$, respectively.

For a continuous function $a : S \rightarrow \mathbb{R}$ with compact support, define

$$\int_S a \, dA = \iint_U a(f(u, v)) \sqrt{\det g(u, v)} \, du \, dv.$$

We must show that using $\tilde{f}$ gives the same answer.

Proof. First, the coordinate-change formula gives

$$\tilde{g}(q) = D\phi(q)^\top g(\phi(q)) D\phi(q).$$

Taking determinants and then positive square roots,

$$\det \tilde{g}(q) = (\det D\phi(q))^2 \det g(\phi(q)),$$

$$\sqrt{\det \tilde{g}(q)} = |\det D\phi(q)| \sqrt{\det g(\phi(q))}.$$

Thus the area factor acquires exactly the absolute Jacobian determinant needed for a change of variables.

Write $x = (u, v)$, $\underline{q} = (r, s)$, and $x = \phi(q)$, with $dx = du dv$ and $dq = dr ds$. Using $\tilde{f} = f \circ \phi$ and the area-factor formula,

$$\begin{aligned} & \iint_{\tilde{U}} a(\tilde{f}(q)) \sqrt{\det \tilde{g}(q)} dq \\ &= \iint_{\tilde{U}} a(f(\phi(q))) \sqrt{\det g(\phi(q))} |\det D\phi(q)| dq \\ &= \iint_U a(f(x)) \sqrt{\det g(x)} dx. \end{aligned}$$

The last equality is the ordinary change-of-variables theorem for double integrals. This proves the lemma.

The function values and the area element must both be transformed:

$$\begin{aligned} a \circ \tilde{f} &= (a \circ f) \circ \phi, \\ \sqrt{\det \tilde{g}(q)} dq &= \sqrt{\det g(\phi(q))} |\det D\phi(q)| dq. \end{aligned}$$

The absolute value is essential: reversing the orientation of the parameters does not change a scalar surface integral.

The same argument applies to any integrable a , and on any overlap of surface coordinate patches. Taking $a = 1$ also proves that the area of a surface region is independent of its parametrization.

Let $S \subset \mathbb{R}^3$ be a smooth surface. A smooth **vector field along** S assigns a vector to each point:

$$X : S \longrightarrow \mathbb{R}^3, \quad p \longmapsto X(p).$$

Smoothness means that $X \circ f$ is smooth in every local parametrization f . The vector $X(p)$ is thought of as based at p , but need not lie in its tangent plane.

The field is **tangent** if $X(p) \in T_p S$ for every p . Locally, this means

$$X(f(u, v)) = a(u, v)f_u + b(u, v)f_v$$

for smooth functions a, b . A vector field *on* S usually means a tangent vector field.

A field N is **normal** if

$$\langle N(p), v \rangle = 0 \quad \text{for every } v \in T_p S.$$

It is a **unit normal field** if also $\|N(p)\| = 1$. Tangent and normal fields may vanish; a unit normal field cannot.

On the unit sphere, for example,

$$X(x, y, z) = (-y, x, 0), \quad N(x, y, z) = (x, y, z).$$

Here X is tangent since $\langle X(p), p \rangle = 0$, whereas N is a unit normal. The tangent field X vanishes at the poles.

An **orientation** of a two-dimensional vector space is a choice of which ordered bases are positive: two bases have the same orientation when their change-of-basis matrix has positive determinant. Exchanging the two basis vectors reverses the orientation.

A surface is **orientable** if we can choose orientations of all its tangent planes consistently as the point varies. Precisely, we require local parametrizations covering S whose coordinate changes $\tilde{f} = f \circ \phi$ satisfy

$$\det D\phi > 0 \quad \text{on every overlap.}$$

Then the ordered bases (f_u, f_v) give compatible orientations. An **oriented surface** is an orientable surface together with a particular choice of orientation.

Every surface patch has a smooth unit normal field

$$N_f(f(u, v)) = \frac{f_u \times f_v}{\|f_u \times f_v\|}.$$

The denominator is nonzero because f is an immersion. Reversing the order of the parameters changes N_f to $-N_f$.

The issue is global: can these local choices be made to agree on all overlaps? Local unit normals always exist; a global one need not.

Theorem. A smooth surface $S \subset \mathbb{R}^3$ is orientable if and only if it admits a smooth unit normal field $N : S \rightarrow \mathbb{R}^3$.

Proof sketch. If $\tilde{f}(r, s) = f(\phi(r, s))$, the chain rule and bilinearity of the cross product give, at corresponding points,

$$\tilde{f}_r \times \tilde{f}_s = \det D\phi (f_u \times f_v).$$

Consequently the associated local unit normals satisfy

$$N_{\tilde{f}} = \frac{\det D\phi}{|\det D\phi|} N_f.$$

For an oriented collection of patches, all these determinants are positive. The local unit normals therefore agree on overlaps and define a smooth unit normal field on S .

Conversely, given N , declare a basis (v, w) of $T_p S$ positive when

$$\langle v \times w, N(p) \rangle > 0.$$

On each sufficiently small connected patch, $N_f = N$ or $N_f = -N$. If necessary, exchange its parameters to obtain $N_f = N$. The displayed transformation law then forces all transition determinants to be positive. Thus S is orientable.

A continuous unit normal field suffices: locally it equals $\pm N_f$, with a locally constant sign, so it is automatically smooth.

On the sphere S_R^2 , the fields

$$N(p) = \frac{p}{R} \quad \text{and} \quad -N(p) = -\frac{p}{R}$$

give the outward and inward orientations. For the cylinder $x^2 + y^2 = R^2$, a global unit normal is

$$N(x, y, z) = \frac{(x, y, 0)}{R}.$$

More generally, if a surface is a regular level set of a *single* smooth function on an open subset of $\mathbb{R}^3$,

$$S = F^{-1}(0), \quad \nabla F \neq 0 \text{ on } S,$$

then it is orientable. Indeed, $T_p S = \ker dF_p$, so

$$N(p) = \frac{\nabla F(p)}{\|\nabla F(p)\|}$$

is a global smooth unit normal field. Separate local defining functions do not by themselves give a consistent global normal.

On a connected orientable surface there are exactly two choices of unit normal, N and $-N$: any other choice equals $\varepsilon(p)N(p)$, where the continuous function $\varepsilon : S \rightarrow \{1, -1\}$ must be constant.

Orientability concerns a *normal* field, not a nowhere-vanishing tangent field. We can now choose a unit normal and study how it changes along the surface; this will lead to the second fundamental form.